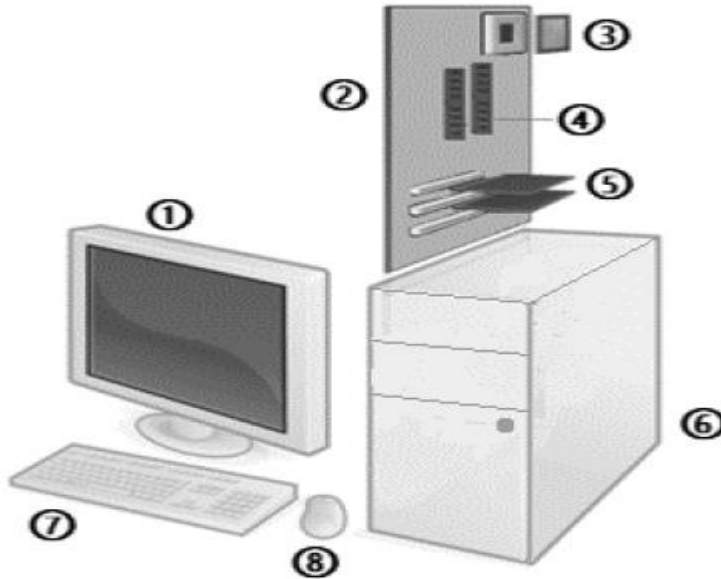


Module: Electronics, system components

Series of TD n° 03

Exercise 1:

Name the following computer components:



Exercise 2:

- 1) Where is the motherboard located? What is its function?
- 2) Name the main components of the motherboard?
- 3) What are the Motherboard Features?
- 4) What does the size factor of a motherboard mean?
- 5) What is the role of the chipset?
- 6) The chipset is often broken down into 2 chips (the Northbridge and the SouthBridge). What is the chip that manages the dialogues between the processor and the RAM?

Exercise 3:

1) It is possible to start a computer without a motherboard

A- True B- False

2) The element that connects all the elements constituting a computer is:

- A. The chipset
- B. The central unit
- C. The motherboard
- D. The processor

3) The role of the processor in a computer is: (choose the correct answers)

- A. To transfer the information
- B. To carry out the instructions
- C. To order all units
- D. To store the instructions
- E. To control the progress of all operations

4) The command-and-control unit

- A. Determines the operation to be executed by the UAL using the sequencer
- B. Completes reading instructions
- C. Performs arithmetic and logical instructions
- D. Produces orders

5) Instruction register stores the address of the instruction being executed

- A- True B- False

6) The state register:

- A. Allows you to store indicators on the state of the system (withholding, overflow, etc.).
- B. Receives the instruction that must be executed.
- C. Determines which operation must be performed, among all possible operations.
- D. Generates control signals.

7) The accumulator

- A. Stores the result of the currently executed instruction
- B. Stores the address of the instruction currently being executed
- C. Stores the instruction currently being executed
- D. Determines the operation to be carried out by the UAL

8) Indicate the correct proposition among the following:

- A. The UAL is responsible for controlling the transfer of data and instructions from central memory to the processor.
- B. The control unit determines the operation to be executed by the UAL using the sequencer.
- C. The clock generates the control signals.
- D. The UAL performs logical operations and arithmetic operations.

9) The arithmetic and logic unit are essentially composed of:

- A. Arithmetic and Logic Unit, Ordinal Counter, Decoder
- B. Ordinal counter, Decoder, Sequencer
- C. Arithmetic and Logic Unit, State Register, Accumulators
- D. Status register, Clock, Sequencer

10) MIPS is a unit of measurement:

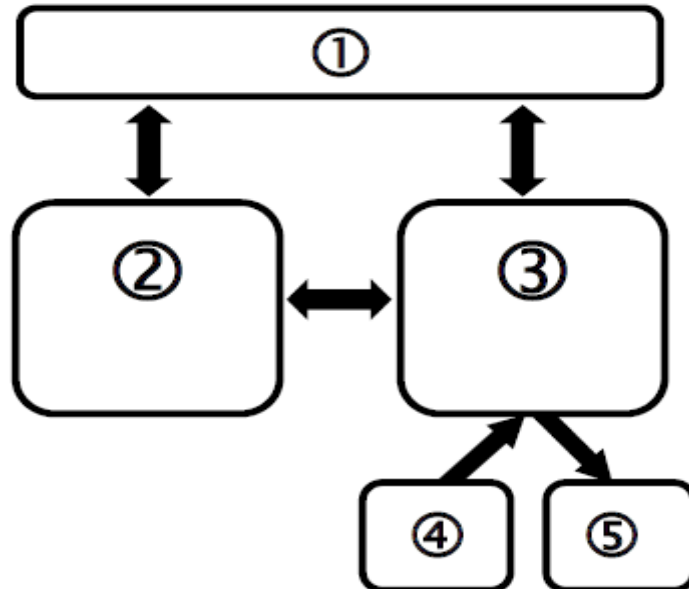
- A. Capacity: Millions of Inches Per System
- B. Calculation speed: Mega Integer Per Switch
- C. Power: Millions of Instructions Per Second
- D. Integration designed by the MIPS company

Exercise 4:

- 1) Convert the following capacities to byte: 13 MB, 2.3 GB, 1.435 TB, 123 KB
- 2) Convert the following capacities to MB: 13 GB, 9 TB, 1400 KB, 23 KB, 1,435,897 o

Exercise 5:

1) Name the parts of Von Neumann's architecture:



2) Give the definition of a processor? What is his role?

3) Name the two main processor manufacturers? give a model for each manufacturer?

4) Why is a processor topped with a heat sink?

5) What are the different components of the processor? What is the role of each component?

6) Classify the following memories by size, by speed: RAM, registers, hard disks, L1 cache, L2 cache, CD-ROM. What do you notice?

7) Specify the function of RAM in a computer. What are its two main characteristics?

8) What are the two types of RAM memory? What are the main differences between them? Where do we use each?

9) Read-only memory comes in different forms which differ essentially in the way they are erased. Mention these forms when explaining the principle of erasing data.

Good luck
Dr. Nadjat Azzaoui